

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-II]

TEST # 06

DATE : 19-01-2020

REVISION TEST

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

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1. A uniform circular disc A of radius r is made from a copper plate of thickness t and another uniform circular disc B of radius $2r$ is made from a copper plate of thickness $t/2$. The relation between the moments of inertia I_A and I_B is

- (1) $I_A > I_B$
(2) $I_A = I_B$
(3) $I_A < I_B$
(4) depends on the values of t and r .

2. The instantaneous angular position of a point on a rotating wheel is given by the equation :-

$$\theta = 2t^3 - 6t^2$$

The torque on the wheel becomes zero at

- (1) $t = 0.5$ sec (2) $t = 0.25$ sec
(3) $t = 2$ sec (4) $t = 1$ sec

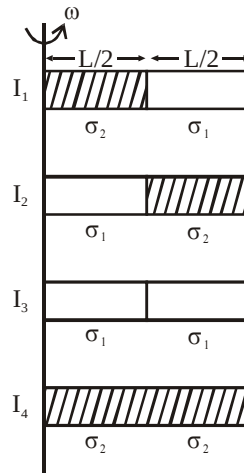
3. The radius of gyration of a rod of length L and mass M about an axis perpendicular to its length and passing through a point at a distance $L/3$ from one of its ends is :-

- (1) $\frac{\sqrt{7}}{6}L$ (2) $\frac{L}{9}$
(3) $\frac{L}{3}$ (4) $\frac{\sqrt{5}}{2}L$

4. We have two spheres one of which is hollow and the other solid. They have identical masses and moment of inertia about their respective diameters. The ratio of their radius is given by:-

- (1) $5 : 7$ (2) $3 : 5$
(3) $\sqrt{3} : \sqrt{5}$ (4) $\sqrt{3} : \sqrt{7}$

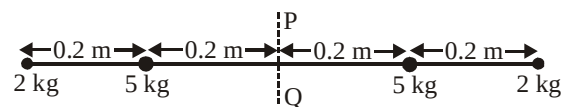
5.



If all rods have same length and area of cross-section. Also density $(\sigma_2) >$ density (σ_1) then :-

- (1) $I_4 > I_2 > I_3 > I_1$
(2) $I_4 > I_3 > I_2 > I_1$
(3) $I_1 > I_2 > I_3 > I_4$
(4) $I_4 > I_2 > I_1 > I_3$

6. Four masses are fixed on a massless rod as shown in the figure. The moment of inertia about the axis PQ is about :-



- (1) 2 kg-m^2
(2) 1.04 kg-m^2
(3) 0.5 kg-m^2
(4) 0.3 kg-m^2

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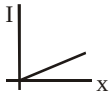
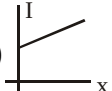
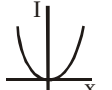
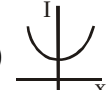
7. The moment of inertia of a solid sphere about an axis passing through centre of gravity is $\frac{2}{5} MR^2$; then its radius of gyration about a parallel axis at a distance $2R$ from first axis is:-

- (1) $5R$ (2) $\sqrt{22/5}R$
(3) $\frac{5}{2}R$ (4) $\sqrt{12/5}R$

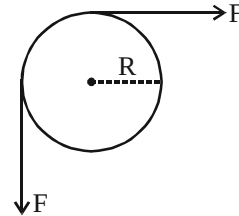
8. Axis X and Z lying in the plane of the disc are mutually perpendicular and axis Y is perpendicular to the plane of the disc. If the moment of inertia of the disc about the axes X and Y are 20 kg-m^2 and 40 kg-m^2 respectively then the moment of inertia of the disc about Z-axis would be

- (1) 60 kg-m^2 (2) 40 kg-m^2
(3) 20 kg-m^2 (4) Zero

9. Figure represents the moment of inertia of the solid sphere about an axis parallel to the diameter of the solid sphere and at a distance x from it. Which one of the following represents the variation of I with x :-

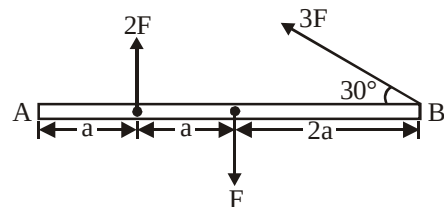
- (1)  (2) 
(3)  (4) 

10. Two equal forces are applied tangentially to a uniform disc of mass M and radius R as shown in the figure. If the disc is pivoted at its centre and free to rotate in its plane, the angular acceleration of the disc is :-



- (1) $\frac{F}{MR}$ (2) $\frac{\sqrt{2}F}{MR}$
(3) $\frac{2F}{MR}$ (4) Zero

11. Three forces F , $2F$ and $3F$ act on a rod AB which is pivoted at A . The anticlockwise moment of forces F , $2F$ and $3F$ about the pivot are respectively :-



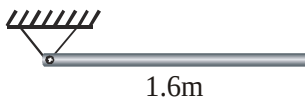
- (1) $+2Fa$; $-3Fa$; $+6Fa$
(2) $-2Fa$; $+2Fa$; $+6Fa$
(3) $-2Fa$; $-2Fa$; $-6Fa$
(4) $+2Fa$; $-2Fa$; $+6Fa$

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12. What is the torque of force $\vec{F} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ acting at a point $\vec{r} = 3\hat{i} + 2\hat{j} + 3\hat{k}$ about the origin?

- (1) $6\hat{i} - 6\hat{j} + 12\hat{k}$ (2) $-6\hat{i} + 6\hat{j} - 12\hat{k}$
(3) $17\hat{i} - 6\hat{j} - 13\hat{k}$ (4) $-17\hat{i} + 6\hat{j} + 13\hat{k}$

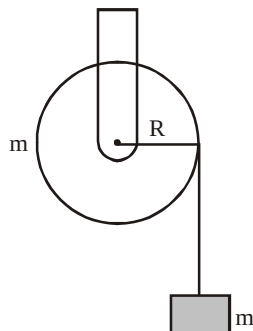
13. The uniform rod of mass 20 kg and length 1.6 m is pivoted at its end and swings freely in the vertical plane. Angular acceleration of rod just after the rod is released from rest in the horizontal position as shown in figure is



- (1) $\frac{15g}{16}$ (2) $\frac{17g}{16}$ (3) $\frac{16g}{15}$ (4) $\frac{g}{15}$

14. A mass 'm' is supported by a massless string wound around a uniform hollow cylinder of mass m and radius R. If the string does not slip on the cylinder, with what acceleration will the mass fall on release?

- (1) $\frac{5g}{6}$
(2) g
(3) $\frac{2g}{3}$
(4) $\frac{g}{2}$



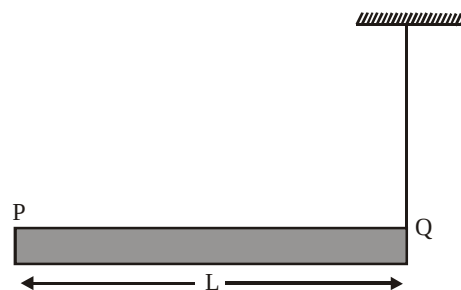
15. A particle is rotating along a circular path in the X-Y plane. The angular momentum vector of the particle will be directed parallel to :-

- (1) x-axis
(2) y-axis
(3) z-axis
(4) none of these

16. Two bodies have their moments of Inertia I and 2I respectively about their axis of rotation. If their kinetic energies of rotation are equal, their angular momentum will be in the ratio :-

- (1) 1:2
(2) $\sqrt{2}:1$
(3) $1:\sqrt{2}$
(4) 2:1

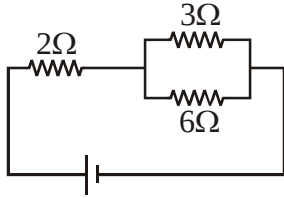
17. A rod PQ of mass M and length L is hinged at end P. The rod is kept horizontal by a massless string tied to point Q as shown in figure. When string is cut, the initial angular acceleration of the rod is :-



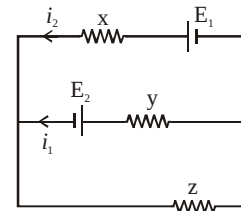
- (1) g/L (2) $2g/L$
(3) $\frac{2g}{3L}$ (4) $\frac{3g}{2L}$

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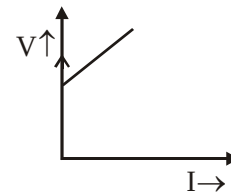
18. A current of 2A flows through 6Ω resistance shown in figure. The power dissipated in the 2Ω resistor is :-



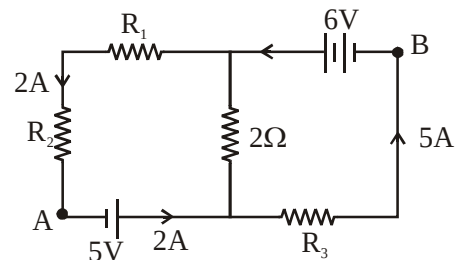
- (1) 36 W (2) 18 W
(3) 72 W (4) 144 W
19. A Galvanometer of resistance 100Ω is connected to a battery of 5V along with a resistance of 4900Ω in series. A full scale deflection of 40 division is obtained in galvanometer. In order to reduce this deflection to 20 division, the additional resistance in series should be :-
(1) 1000Ω (2) 9900Ω
(3) 5000Ω (4) 10000Ω
20. The product of resistivity and conductivity of material is :-
(1) zero (2) Infinity
(3) unity (4) None
21. An electric kettle takes 2A current at 200 V. Then how much time will it take to boil 2kg of water from temperature 40°C . The temperature of boiling water is 100°C :-
(1) 7 min (2) 14 min
(3) 21 min (4) 35 min
22. For the circuit shown in figure, which one of the following equation is correct :-



- (1) $E_1 - i_2x + E_2 + i_1y = 0$
(2) $E_1 - i_2x - i_1z = 0$
(3) $E_2 + i_2y - i_2z = 0$
(4) $-E_2 + (i_1 + i_2)z + i_1y = 0$
23. For the given figure, graph is drawn in between terminal potential difference (V) of a cell of emf E and of internal resistance r as function of current (I), flowing through it, during charging of the cell. The slope of graph and intercept are respectively :-



- (1) $-E$ and r (2) E and r
(3) $-r$ and E (4) r and E
24. In the circuit shown in the figure, if the potential at point A is taken to be -2V , the potential at point B is :-



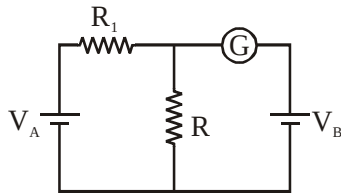
- (1) -3V (2) 3V (3) -4V (4) 5V

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25. A galvanometer of resistance G is shunted by a resistance S ohm. To keep the main current half to initial value in the galvanometer, the resistance to be put in series with the galvanometer is :-

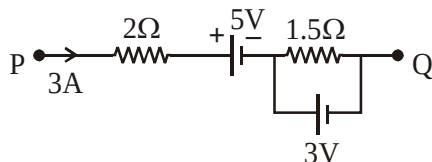
(1) $\frac{G(G+2S)}{(G+S)}$ (2) $\frac{G(2G+S)}{(G+S)}$
(3) $\frac{S(G+2S)}{(G+S)}$ (4) $\frac{S(2G+S)}{(G+S)}$

26. In the circuit shown the cells A and B have negligible resistances. for $V_A = 16$ V, $R_1 = 600 \Omega$ and $R = 200 \Omega$, the galvanometer shows no deflection. The value V_B is :-



- (1) 12 V (2) 4 V (3) 16 V (4) 8 V

27. The potential difference ($V_P - V_Q$) between points P and Q in the given figure is :-

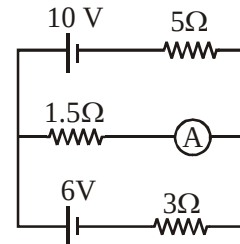


- (1) 7 V (2) 21 V
(3) 14 V (4) 28 V

28. A potentiometer is preferred over voltmeter while measuring voltage because :-

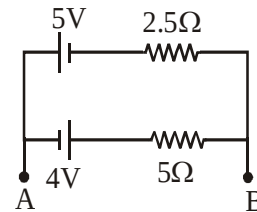
- (1) It doesn't draw any current from main circuit
(2) its resistance is high
(3) its resistance is low
(4) can't be predicted.

29. In the given circuit, reading of ammeter is nearly:-



- (1) 8 A (2) 5 A (3) 1 A (4) 2 A

30. In the given figure if the circuit is reduced into a single cell then emf and internal resistance of equivalent cell is :-



- (1) $\frac{14}{3}$ V, 7.5Ω (2) $\frac{14}{3}$ V, $\frac{5}{3} \Omega$
(3) 2V, $\frac{5}{3} \Omega$ (4) 2V, 7.5Ω

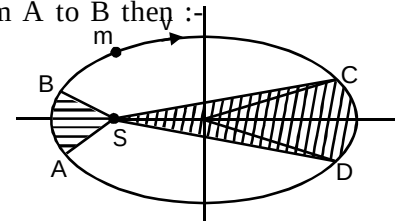
31. A cell of 5V emf and 0.75Ω internal resistance delivers a current of 3A in an external load then the terminal potential difference of the cell is :-

- (1) 2.75 V (2) 7.25 V
(3) 3.5 V (4) 5.75 V

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32. Resistance per unit length of potentiometer wire is $10^{-4} \Omega/\text{m}$ and current flowing through it is 10A, then the potential gradient will be :-
 (1) 0.1 mV/m (2) 10 mV/m
 (3) 1 mV/m (4) 100 mV/m
33. For a satellite moving in an orbit around the earth, the ratio of potential energy to kinetic energy is:-
 (1) $\frac{1}{\sqrt{2}}$ (2) 2 (3) $\sqrt{2}$ (4) $\frac{1}{2}$
34. An artificial satellite moving in a circular orbit at a distance h from the centre of the earth has total energy E_0 . Its potential energy is :-
 (1) E_0 (2) $\frac{E_0}{2}$
 (3) $2E_0$ (4) $1.5 E_0$
35. Gravitational acceleration on the surface of a planet is $\frac{\sqrt{6}}{11}g$, where g is the acceleration on the surface of the earth. The average mass density of the planet is $\frac{2}{3}$ times that of earth. If the escape speed on the surface of the earth is 11 km/s, the escape speed on the surface of the planet will be :-
 (1) 3 km/s (2) 6 km/s
 (3) 2 km/s (4) 9 km/s
36. The Earth is assumed to be a sphere of radius R . A platform is arranged at a height $2R$ from the surface of the Earth. The escape velocity of a body from this platform is fv , where v is its escape velocity from the surface of the Earth. The value of f is :-

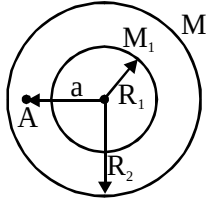
- (1) $\frac{1}{\sqrt{3}}$ (2) $\sqrt{3}$ (3) 3 (4) $\frac{1}{3}$
37. The time period of a satellite in a circular orbit around a planet is independent of
 (1) The mass of the planet
 (2) The radius of the orbit
 (3) The mass of the satellite
 (4) All of the above
38. The figure shows elliptical orbit of a planet m about the sun S . The shaded area SCD is thrice the shaded area SAB . If t_1 is the time for the planet to move from C to D and t_2 is the time to move from A to B then :-
 (1) $t_1 = t_2$
 (2) $t_1 < t_2$
 (3) $t_1 = 4t_2$
 (4) $t_1 = 3t_2$
39. The radii of circular orbits of two satellites A and B of the earth, are $9R$ and R , respectively. If the speed of satellite A is V , then the speed of satellite B will be :-
 (1) V (2) $3V$
 (3) $\frac{V}{3}$ (4) $V\sqrt{3}$
40. A particle of mass M is situated at the centre of a spherical shell of same mass and radius r . The gravitational potential at a point situated at $\frac{r}{4}$ distance from the centre, will be :-



- (1) $-\frac{GM}{4}$ (2) $-\frac{3GM}{r}$
 (3) $-\frac{4GM}{r}$ (4) $-\frac{5GM}{r}$

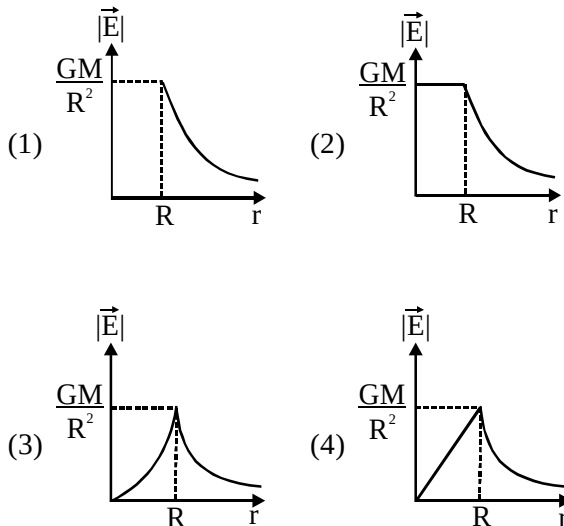
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41. The figure represents two concentric shells of radii R_1 and R_2 and masses M_1 and M_2 respectively. The gravitational field intensity at point A at a distance "a" from the centre is



- (1) $\frac{G(M_1 + M_2)}{a^2}$ (2) $\frac{GM_1}{a^2} + \frac{GM_2}{R_2^2}$
(3) $\frac{GM_1}{a^2}$ (4) Zero

42. Gravitational field due a uniform spherical shell of radius R and mass M varies with r (where r is distance from centre of shell) correctly shown by:-



43. The height at which acceleration due to gravity will become $\frac{1}{4}$ th of its value on the surface of earth. (R is radius of earth)

- (1) R
(2) $2R$
(3) $3R$
(4) $4R$

44. A satellite-1 is orbiting the earth at a height R above surface of the earth. R is being radius of the earth. If its time period is T . Satellite-2 which is orbiting the earth at a height $7R$ then its time period will be :-

- (1) $8T$
(2) $\frac{T}{8}$
(3) $4T$
(4) $\frac{T}{4}$

45. The mass of planet mars is M and orbital radius of its moon is R . The time period of moon of mars is

- (1) $T^2 = \frac{4\pi^2 R^3}{GM}$
(2) $T^2 = \frac{4\pi^2 GR^3}{M}$
(3) $T^2 = \frac{2\pi R^3 G}{M}$
(4) $T^2 = 4\pi MGR^3$

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46. Select the rate law that corresponds to the data shown for the reaction $A + B \rightarrow C$

Exp.	[A]	[B]	Initial Rate
1.	0.012	0.035	0.10
2.	0.024	0.070	0.80
3.	0.024	0.035	0.10
4.	0.012	0.070	0.80

- (1) Rate = $k [B]^3$
 (2) Rate = $k [B]^4$
 (3) Rate $k = [A] [B]^3$
 (4) Rate = $k [A]^2 [B]^2$

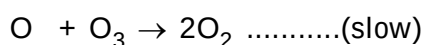
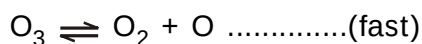
47. The activation energy for the forward reaction $X \rightarrow Y$ is 60. KJ mol^{-1} and ΔH is -20 KJ mol^{-1} , the activation energy for the reverse reaction is -

- (1) 40 KJ mol^{-1}
 (2) 60 KJ mol^{-1}
 (3) 80 KJ mol^{-1}
 (4) 20 KJ mol^{-1}

48. In a first order reaction, 75% of the reactant disappeared in 1.386 hours. Calculate the rate constant of the reaction -

- (1) $3.6 \times 10^{-3} \text{ sec}^{-1}$
 (2) $2.8 \times 10^{-4} \text{ sec}^{-1}$
 (3) $7.2 \times 10^{-3} \text{ sec}^{-1}$
 (4) $1.8 \times 10^{-3} \text{ sec}^{-1}$

49. The chemical reaction $2\text{O}_3 \rightarrow 3\text{O}_2$ proceeds as follows -



The rate law expression should be -

- (1) $r = k [\text{O}_3]^2$
 (2) $r = k [\text{O}_3]^2 [\text{O}_2]^{-1}$
 (3) $r = k [\text{O}_3] [\text{O}_2]$
 (4) Unpredictable

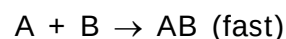
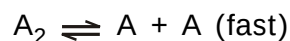
50. The half life period for a certain first order reaction is 30 minutes. How long will it take for $1/32$ of the reactant to be left behind -

- (1) 60 Min. (2) 120 Min.
 (3) 90 Min. (4) 150 Min.

51. As the energy of activation increases -

- (1) The number of reacting molecules colliding decreases
 (2) The rate of reaction decreases
 (3) The rate of reaction increases
 (4) The rate constant increases

52. A hypothetical reaction $A_2 + B_2 \rightarrow 2AB$ follows the mechanism as given below -



The order of the over all reaction is -

- (1) 2 (2) 1 (3) $\frac{3}{2}$ (4) 0

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53. The half life period for catalyst decomposition of AB_3 at 50 mm is found to be 4 hrs and at 100 mm it is 2.0 hrs. The order of reaction is -
(1) 3 (2) 1 (3) 2 (4) 0
54. If the rate of reaction becomes 2 times for every 10°C rise in temperature, by what factor the rate of reaction increases when temperature is increased from 30°C to 80°C
(1) 16 (2) 32 (3) 64 (4) 128
55. If the concentration is expressed in moles per litre the unit of the rate constant for a first order reaction is-
(1) $\text{Mol litre}^{-1} \text{ sec}^{-1}$ (2) Mol litre^{-1}
(3) sec^{-1} (4) Mol sec^{-1}
56. Which of the following represents the expression for $3/4$ th life of first order reaction
(1) $\frac{k}{2.303} \log 4/3$ (2) $\frac{2.303}{k} \log 3/4$
(3) $\frac{2.303}{k} \log 4$ (4) $\frac{2.303}{k} \log 3$
57. For the reaction $A \rightarrow B$, data of initial concentration and corresponding half life period are given in the tabular form
- | [A] | 1M | 2M | 4M |
|-----------|------|------|-------|
| $T_{0.5}$ | 300s | 600s | 1200s |
- The order of the reaction is -
(1) 0 (2) 1 (3) 2 (4) 3
58. For a reaction
 $AB_5 \longrightarrow AB + 4B$
The rate can be expressed in following ways
 $-\frac{d[AB_5]}{dt} = K[AB_5]$; $\frac{d[B]}{dt} = K_1[AB_5]$
So the correct relation between K and K_1 is-
(1) $K_1 = K$ (2) $K_1 = 2K$
(3) $K_1 = 4K$ (4) $2K_1 = K$
59. Which element is having lowest melting and boiling point :
(1) Ti (2) Cu
(3) Zn (4) Mn
60. To which of the following series the transition element from $Z = 39$ to $Z = 48$ belong :
(1) 3d series (2) 4d series
(3) 5d series (4) 6d series
61. Chromium forms most stable compound in the following oxidation state :
(1) Cr (I) (2) Cr (II)
(3) Cr (III) (4) Cr (IV)
62. $N_2(g) + 3H_2(g) \xrightleftharpoons{\text{Fe + Mo}} 2NH_3(g)$; Haber's process, Mo is used as :
(1) A catalyst
(2) A catalytic promoter
(3) An oxidising agent
(4) As a catalytic poison

SPACE FOR ROUGH WORK

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| <p>63. What is DDT among the following :</p> <p>(1) Greenhouse gas
(2) A fertilizer
(3) Biodegradable pollutant
(4) Non-biodegradable pollutant</p> <p>64. Identify the pollutant gases largely responsible for the discoloured and lustreless nature of marble of the Taj Mahal</p> <p>(1) O_3 and CO_2 (2) CO_2 and NO_2
(3) SO_2 and O_3 (4) SO_2 and NO_2</p> <p>65. As compared to potassium, sodium has</p> <p>(1) Lower electronegativity
(2) Higher ionization potential
(3) Greater atomic radius
(4) Lower melting point</p> <p>66. Which of the following is strongest reducing agent</p> <p>(1) Li (2) K
(3) Na (4) Rb</p> <p>67. The colour given to the flame by sodium salts is</p> <p>(1) Light red (2) Golden yellow
(3) Green (4) Pink</p> <p>68. Photoelectric effect is maximum in</p> <p>(1) Cs (2) Na
(3) K (4) Li</p> <p>69. The sum of protons, electrons and neutrons in the heaviest isotope of hydrogen is</p> <p>(1) 6 (2) 5
(3) 4 (4) 3</p> | <p>70. Temporary hardness may be removed from water by adding</p> <p>(1) $CaCO_3$ (2) $Ca(OH)_2$
(3) $CaSO_4$ (4) HCl</p> <p>71. In which of the following reaction hydrogen peroxide is a reducing agent</p> <p>(1) $2FeCl_2 + 2HCl + H_2O_2 \rightarrow 2FeCl_3 + 2H_2O$
(2) $Cl_2 + H_2O_2 \rightarrow 2HCl + O_2$
(3) $2HI + H_2O_2 \rightarrow 2H_2O + I_2$
(4) $H_2SO_3 + H_2O_2 \rightarrow H_2SO_4 + H_2O$</p> <p>72. Sucrose on hydrolysis gives:</p> <p>(1) glucose only
(2) glucose and galactose
(3) glucose and fructose
(4) glucose and lactose</p> <p>73. The disaccharide present in milk is:</p> <p>(1) sucrose (2) lactose
(3) maltose (4) none of these</p> <p>74. Helical structure of proteins is stabilized by:</p> <p>(1) ionic bond
(2) covalent bond
(3) van der Waals' forces
(4) hydrogen bonds</p> <p>75. Glucose and fructose are:</p> <p>(1) optical isomers
(2) functional group isomers
(3) chain isomers
(4) position isomers</p> |
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SPACE FOR ROUGH WORK

76. Which polymers occur naturally?

- (1) Starch and Nylon
- (2) Starch and Polythene
- (3) Proteins and Nylon
- (4) Proteins and Natural Rubber

77. Which of the following have branched structure?

- (1) Low density polythene
- (2) High density polythene
- (3) Amylose
- (4) Cellulose

78. Which statement about aspirin is not true

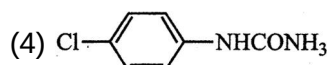
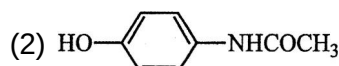
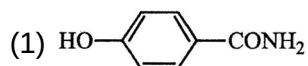
- (1) Aspirin belongs to narcotic analgesics.
- (2) It is effective in relieving pain.
- (3) It has anti blood clotting action.
- (4) It is a neurologically active drug.

79. Which compound/set of compounds is used in the, manufacture of Nylon-6, 6 ?

- (1) $\text{HOOC}(\text{CH}_2)_4\text{COOH} + \text{H}_2\text{N}(\text{CH}_2)_6\text{NH}_2$
- (2) $\text{CH}_3 = \text{CH}-\text{C}(\text{CH}_3) = \text{CH}_2$
- (3) $\text{CH}_2 = \text{CH}_2$

- (4) $\text{HOOC}-\text{C}_6\text{H}_4-\text{COOH} + \text{HOCH}_2-\text{CH}_2\text{OH}$

80. The correct structure of the drug paracetamol is



81. The compound that causes general antidepressant action on the central nervous system belongs to the class of _____

- (1) analgesics
- (2) tranquilizers
- (3) narcotic analgesics
- (4) antihistamines

82. Name the simplest amino acid

- (1) Alanine
- (2) Tyrosine
- (3) Asparagine
- (4) Glycine

83. The process which involves the treatment of the ore with a suitable reagent as to make it soluble while impurities remain insoluble is called :-

- (1) froth floatation process
- (2) leaching
- (3) self reduction
- (4) hydrometallurgy

SPACE FOR ROUGH WORK

84. Which of the reactions occurs in slag formation zone in blast furnace for manufacture of iron ?

 - (1) $\text{Fe}_2\text{O}_3 + 3\text{C} \rightarrow 2\text{Fe} + 3\text{CO}$
 - (2) $\text{CO}_2 + \text{C} \rightarrow 2\text{CO}$
 - (3) $\text{FeO} + \text{SiO}_2 \rightarrow \text{FeSiO}_3$
 - (4) $\text{CaO} + \text{SiO}_2 \rightarrow \text{CaSiO}_3$

85. When ZnS and PbS minerals are present together, Then NaCN added to separate them in the froth floatation process as a depressant, because :-

 - (1) Pb(CN)_2 is precipitated while no effect on ZnS
 - (2) ZnS forms soluble complex $\text{Na}_2[\text{Zn(CN)}_4]$
 - (3) PbS forms soluble complex $\text{Na}_2[\text{Pb(CN)}_4]$
 - (4) It decreases the floatation property of PbS by making it hydrophilic

86. The method of zone refining of metals is based on the principle of :-

 - (1) Greater mobility of the pure metal than that of impurity
 - (2) higher melting point of the impurity than that of the pure metal
 - (3) Greater noble character of the solid metal than that of the impurity
 - (4) Greater solubility of the impurity in the molten state than in the solid

87. Oil used as frothing agent in froth floatation process is

 - (1) Pine oil
 - (2) Mustard oil
 - (3) Coconut oil
 - (4) Olive oil

88. Roasting is generally done in case of

 - (1) Oxide ores
 - (2) Silicate ores
 - (3) Sulphide ores
 - (4) Carbonate ores

89. Use of electrolysis is

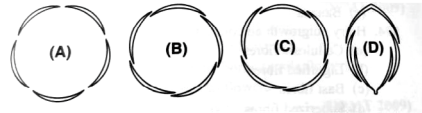
 - (1) Electroplating
 - (2) Electrefining
 - (3) Both (1) and (2)
 - (4) None of these

90. The substance which is added to remove impurities is known as

 - (1) Slag
 - (2) Flux
 - (3) Gangue
 - (4) Catalyst

SPACE FOR ROUGH WORK

91. Drupe has :
 (1) Hard epicarp (2) Hard endocarp
 (3) Hard mesocarp (4) No epicarp
92. A flower which can be divided into equal vertical halves by more than one plane of division is :
 (1) Actinomorphic (2) Zygomorphic
 (3) Heteromorphic (4) Cyclic
93. Replum is present in the ovary of :
 (1) Sunflower (2) Pea
 (3) Lemon (4) Mustard
94. Unilocular ovary with a single ovule, the placentaion is :
 (1) Marginal (2) Basal
 (3) Free central (4) Axil
95. Trimerous flowers, superior ovary and axile placentation are chatacteristics of :
 (1) Liliaceae (2) Cucurbitaceae
 (3) Solanaceae (4) Compositae
96. Coir is obtained from the fruit of coconut from its :
 (1) Epicarp (2) Mesocarp
 (3) Endocarp (4) Pericarp
97. Which of the following floral formula represents the mustard plant ?
 (1) $\oplus \text{ } \overline{\text{K}}_{2+2} \text{C}_4 \text{A}_{2+4} \underline{\text{G}}_{(2)}$
 (2) $\oplus \text{ } \overline{\text{K}}_{2+2} \text{P}_{2+2} \text{C}_4 \text{A}_{2+1} \underline{\text{G}}_{(2)}$
 (3) $\oplus \text{ } \overline{\text{K}}_{(5)} \text{C}_{(5)} \widehat{\text{A}_5} \underline{\text{G}}_{(2)}$
 (4) $\oplus \text{ } \overline{\text{K}}_{2+2} \text{C}_{4 \times} \text{A}_{2+4} \underline{\text{G}}_{(2)}$
98. Consider the following statements :
 A. In racemose inflorescence, flowers are basipetally arranged.
 B. Epigynous flowers are seen in rose plant.
 C. In brinjal, the ovary is superior.
 Of these statements :
 (1) A and B are true but C is false
 (2) A and C are ture but B is false
 (3) A and B are false but C is true
 (4) B and C are true but A is false

99. Ovary is half inferior in :
 (1) Guava (2) Peach
 (3) Brinjal (4) Cucumber
100. The correct floral formula of chilli is :
 (1) $\oplus \text{ } \overline{\text{K}}_{5} \text{C}_{5} \text{A}_{(5)} \underline{\text{G}}_{(2)}$
 (2) $\oplus \text{ } \overline{\text{K}}_{(5)} \text{C}_{(5)} \widehat{\text{A}_5} \underline{\text{G}}_{(2)}$
 (3) $\oplus \text{ } \overline{\text{K}}_{5} \text{C}_{(5)} \text{A}_{(5)} \underline{\text{G}}_{(2)}$
 (4) $\oplus \text{ } \overline{\text{K}}_{(5)} \text{C}_{(5)} \text{A}_{(5)} \underline{\text{G}}_{(2)}$
101. The following diagrams represent the types of aestivation in corolla. Identify the correct combination of labelling :

 (1) A = Valvate, B = Twisted, C = Vexillary, D = Imbricate
 (2) A = Valvate, B = Vexillary, C = Twisted, D = Imbricate
 (3) A = Vexillary, B = Imbricate, C = Twisted, D = Valvate
 (4) A = Valvate, B = Twisted, C = Imbricate, D = Vexillary
102. From the options given below, find out the correct floral formula for a flower having the following characters namely actinomorphic, bisexual, five united sepals, five united petals, stamens five and epipetalous, ovary bicarpellary, syncarpous and superior :
 (1) $\oplus \text{ } \overline{\text{K}}_{(5)} \text{C}_{(5)} \widehat{\text{A}_5} \underline{\text{G}}_2$
 (2) $\oplus \text{ } \overline{\text{K}}_{(5)} \text{C}_{(5)} \widehat{\text{A}_5} \underline{\text{G}}_{(2)}$
 (3) $\oplus \text{ } \overline{\text{K}}_{(5)} \text{C}_{(5)} \text{A}_{(5)} \underline{\text{G}}_{(2)}$
 (4) $\oplus \text{ } \overline{\text{K}}_{(5)} \text{C}_{(5)} \widehat{\text{A}_5} \underline{\text{G}}_{(2)}$
103. Placentation in tomato and lemon is :
 (1) Marginal (2) Axile
 (3) Parietal (4) Free - central

104. Vexillary aestivation is characteristic of the family :

- (1) Solanaceae
- (2) Brassicaceae
- (3) Fabaceae
- (4) Asteraceae

105. The modified stem of *Opuntia* is :

- (1) Phyllode
- (2) Phylloclade
- (3) Cladode
- (4) Staminode

106. Study the following statements and choose the correct option :

- I. Buds are present in the axil of leaflets of the compound leaf.
- II. Pulvinus leaf-base is present in some leguminous plants.
- III. In *Opuntia*, the petioles expand, become green and synthesize food.
- IV. Opposite phyllotaxy is seen in guava, *Calotropis*

Options :

- (1) II and IV are correct but I and III are wrong
- (2) I and III are correct but II and IV are wrong
- (3) I and IV are correct but II and III are wrong
- (4) II, III and IV are correct but I is wrong

107. Inflorescence of *Ficus* is :

- (1) Raceme
- (2) Spike
- (3) Hypanthodium
- (4) Verticillaster

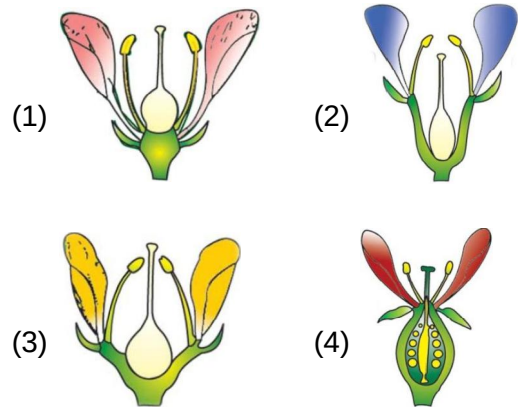
108. The inflorescence in *Ocimum* is :

- (1) Cyathium
- (2) Verticillaster
- (3) Hypanthodium
- (4) Raceme

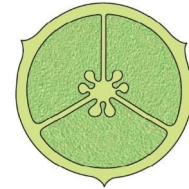
109. Keel is characteristic of the flowers of :

- (1) Gulmohur, rose
- (2) *Cassia*, tomato
- (3) *Calotropis*, tomato
- (4) Bean, pea

110. In which given following diagram which shows superior ovary

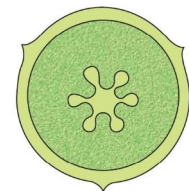


111. According to following diagram select suitable explanation of gynoecium



- (1) Trilocular, Tricarpallary, Apocarpus
- (2) Unilocular, Tricarpallary, Syncarpous
- (3) Trilocular, Tricarpallary, Syncarpous
- (4) Trilocular, Monocarpellary, Syncarpous

112. In given following figure identify type of placentation



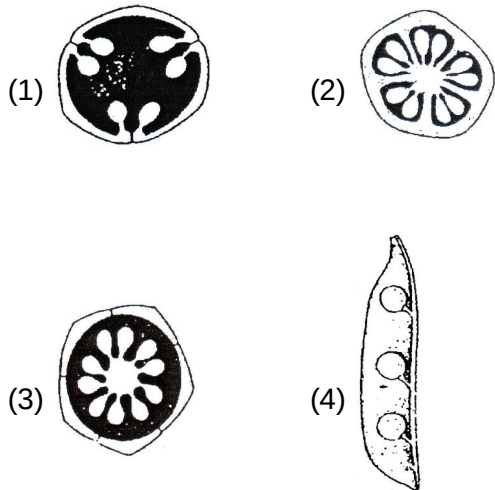
- (1) Axile
- (2) Free central
- (3) Basal
- (4) Superficial

113. Identify this inflorescence :-



- (1) Cymose
- (2) Cyathium
- (3) Racemose
- (4) Hypanthodium

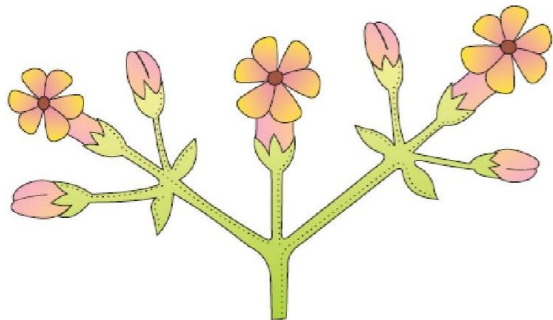
114. Which of the following diagrams represents the type of placentation as described under "When the ovules are borne on central axis and septa are absent"



115. Aestivation of petals in the flower of china rose, lady's finger (Malvaceae) is correctly shown in :-



116.



Choose incorrect statement regarding the above figure given.

- (a) In this inflorescence main axis terminate into a flower.
- (b) There is a limited growth.
- (c) Flowers are borne in a acropetal order.
- (d) Shown by Bougainvillia

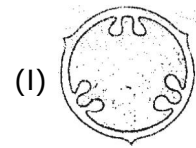
- (1) b, c
- (2) only c
- (3) c, d
- (4) b, c, d

117. Match the Column-I and Column-II :

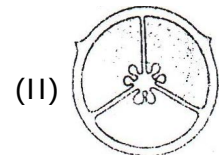
Column-I

Column-II

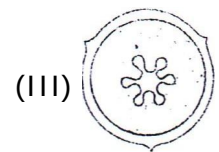
(A) Pea



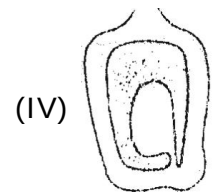
(B) Lemon



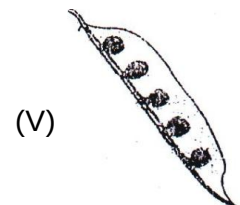
(C) Mustard



(D) Primrose



(E) Sunflower



- (1) A – V, B – II, C – I, D – III, E – IV
- (2) A – I, B – II, C – III, D – IV, E – V
- (3) A – V, B – II, C – III, D – I, E – IV
- (4) A – V, B – III, C – II, D – I, E – IV

118. Restriction endonucleases

- (1) cleave DNA at highly specific recognition sequences
- (2) are inserted into bacteria by bacteriophages
- (3) contain axon and intron
- (4) add methyl groups to specific DNA sequences

119. During isolation of DNA, addition of which of the following causes precipitation of purified DNA ?

- (1) Chilled ethanol
- (2) RNA primer
- (3) DNA polymerase
- (4) Proteases

120. A bioreactor (fermenter) refers to

- (1) a device in which substances are treated to stimulate biochemical transformation by living cells
- (2) study of mutated species
- (3) a tank for biochemical reactions
- (4) organisms reacting to a stimulus

121.

Column-I		Column-II	
I	PCR	A	Large scale culture
II	Bioreactor	B	To induce alien DNA in host cell
III	Genegun	C	Restriction endonuclease
IV	EcoR1	D	Amplification of gene

- (1) I-D, II-A, III-B, IV-C
- (2) I-B, II-A, III-D, IV-C
- (3) I-C, II-B, III-D, IV-A
- (4) I-A, II-D, III-B, IV-C

122.

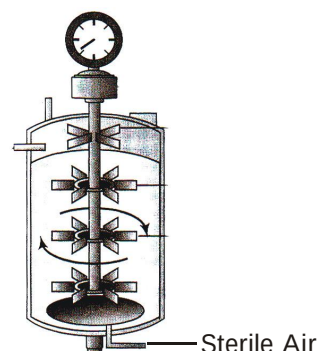
Column-I		Column-II	
I	EcoRI	A	<i>Bacillus amyloliquefaciens</i>
II	Bam HI	B	<i>Haemophilus influenza</i>
III	Hind III	C	<i>Escherichia coli</i>
IV	pBR322	D	Artificial plasmid

- (1) I-C, II-A, III-B, IV-D
- (2) I-B, II-A, III-D, IV-C
- (3) I-D, II-A, III-C, IV-B
- (4) I-A, II-D, III-B, IV-C

123. Select the correct option which shows the most appropriate temperatures of three different steps of PCR mechanism

Denaturation	Primer annealing	Primer extension
(1) 40–60°C	72°C	90°C
(2) 60–60°C	96°C	40–60°C
(3) 94–96°C	40–60°C	72°C
(4) 27°C	45–50°C	85–87°C

124. Identify the correct match for the given apparatus.



Apparatus		Function	
(1)	Genegun	i	Vectorless direct gene transfer
(2)	Paper chromatography	ii	Separation of chlorophyll pigments
(3)	Stirred tank bioreactor	iii	Carry out fermentation process
(4)	Respirometer	iv	Finding out rate of respiration

125. Which of the following statement is not true ?

- (1) Eco RI always cuts DNA molecules at a particular point by recognizing a specific sequence of 4 base pairs
- (2) Besides Hind-II, today we know more than 900 restriction enzymes
- (3) restriction enzymes are part of defence mechanism
- (4) Type-II restriction endonuclease is most useful in genetic engineering

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| <p>126. Antibiotic resistance gene present of Bam HI site of a <i>E.coli</i> cloning vector is</p> <ol style="list-style-type: none"> (1) streptomycin resistance (2) tetracycline resistance (3) chloramphenicol resistance (4) kanamycin resistance <p>127. In vector pBR 322, the 'rop' codes for the</p> <ol style="list-style-type: none"> (1) Pvu II (2) Pvu I (3) Proteins that involved in the replication of the restriction enzyme (4) Proteins that involved in the replication of the plasmid <p>128. Restriction enzymes belongs to a large class of the enzyme called</p> <ol style="list-style-type: none"> (1) Cellulase (2) Nuclease (3) Chitinase (4) Polymerase <p>129. In pBR 322, how many recognition sites are present?</p> <ol style="list-style-type: none"> (1) 8 (2) 7 (3) 3 (4) 4 <p>130. A method of cellular defence that takes place in all eukaryotic organisms is</p> <ol style="list-style-type: none"> (1) RNAi (2) downstream processing (3) extension (4) annealing | <p>131. VNTRs are</p> <ol style="list-style-type: none"> (1) Variable Number Tandem Repeats (2) Very Narrow Tandem Repeats (3) Variable Non-cistronic Transposon Repeats (4) Valuable Non-cistronic Transposic Regions <p>132. You can see bright orange coloured bands of DNA in a ethidium bromide stained gel exposed to</p> <ol style="list-style-type: none"> (1) X-rays (2) sun rays (3) UV rays (4) gamma rays <p>133. The crops start wilting if excess fertilizers are added to the field. This is due to</p> <ol style="list-style-type: none"> (1) imbibition (2) plasmolysis (3) osmosis (4) none of these <p>134. For a fully turgid cell</p> <ol style="list-style-type: none"> (1) O.P. = D.P.D. (2) O.P. = Zero (3) D.P.D. = Zero (4) T.P. = Zero <p>135. A cell becomes turgid when placed in</p> <ol style="list-style-type: none"> (1) isotonic solution (2) hypertonic solution (3) hypotonic solution (4) none of the above |
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| <p>136. Mark the odd one w.r.t. origin.</p> <ol style="list-style-type: none"> (1) Astrocyte (2) Microgliocytes (3) Neuron (4) Oligodendrocyte <p>137. The part of the brain which is most developed in higher primates is :</p> <ol style="list-style-type: none"> (1) cerebellum (2) cerebrum (3) medulla oblongata (4) olfactory lobes <p>138. A mass of cytons is called as :</p> <ol style="list-style-type: none"> (1) synapse (2) ganglion (3) motor nerve (4) nerve fibre <p>139. Neurotransmitter between neuron and a muscle cell is-</p> <ol style="list-style-type: none"> (1) Dopamine (2) Serotonin (3) GABA (4) Acetylcholine <p>140. Which of the following is mismatched-</p> <ol style="list-style-type: none"> (1) Cerebrum-Memory (2) Cerebellum-Equilibrium of body (3) Olfactory lobes - Smell (4) Medulla - Temperature <p>141. If the corpus callosum is removed in mammalian brain then what will be affected:-</p> <ol style="list-style-type: none"> (1) Coordination of cerebrum (2) Involuntary activity of brain (3) Coordination of cerebellum (4) Behaviour of cerebellum <p>142. Acetylcholine is:</p> <ol style="list-style-type: none"> (1) Toxin (2) Vitamin (3) Enzyme (4) Chemical transmitter <p>143. If both electrodes of cathode ray oscilloscope (CRO) are placed inside an axon, then it shows a membrane potential of ____ in resting state.</p> <ol style="list-style-type: none"> (1) -70 mV (2) -55 mV (3) +30 mV (4) 0 mV | <p>144. Nervous band connecting the two cerebral hemispheres is:</p> <ol style="list-style-type: none"> (1) Corpus albicans (2) Corpus callosum (3) Corpus striatum (4) Corpus spongiosum <p>145. Subarachnoid space occurs</p> <ol style="list-style-type: none"> (1) Above piamater (2) Above durameter (3) Inside blastocoele (4) None of the above <p>146. Hypothalamus is:</p> <ol style="list-style-type: none"> (1) Helpful in sleep (2) Has centres for thirst and hunger (3) Controls body temperature (4) All the above <p>147. Third, sixth and eleventh cranial nerves are-</p> <ol style="list-style-type: none"> (1) Oculomotor, Abducens and Accessory spinal (2) Oculomotor, Trigeminal and Accessory spinal (3) Optic, Facial and Accessory spinal (4) Trochlear, Abducens and Vagus <p>148. What is the effect of parasympathetic nervous system</p> <ol style="list-style-type: none"> (1) Originates heart beat (2) Decrease heart beat (3) Increases heart beat (4) No effect on H.B. <p>149. Fifth cranial nerve is</p> <ol style="list-style-type: none"> (1) Optic (2) Trigeminal (3) Abducens (4) Facial <p>150. Brain ventricles are lined by-</p> <ol style="list-style-type: none"> (1) Neurons (2) Schwann cells (3) Neuroglia (4) Ependymal cells |
|--|---|

- | | |
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| <p>151. Meninges form outer envelope in-</p> <p>(1) Cartilage (2) Muscle</p> <p>(3) Brain (4) Kidney</p> <p>152. Brain originates from-</p> <p>(1) Endoderm</p> <p>(2) Mesoderm</p> <p>(3) Ectoderm</p> <p>(4) Ecto-endodermal</p> <p>153. Hypophyseal portal system occurs in-</p> <p>(1) Brain (2) Kidney</p> <p>(3) Liver (4) Heart</p> <p>154. Nissl granules of neuron are made up of-</p> <p>(1) Proteins</p> <p>(2) Mitochondria</p> <p>(3) Ribonucleo protein</p> <p>(4) DNA</p> <p>155. The nerve secreting epinephrine or norepinephrine is called-</p> <p>(1) Adrenergic</p> <p>(2) Cholinergic</p> <p>(3) Ammonelic</p> <p>(4) None of them</p> <p>156. Cerebrospinal fluid is formed by-</p> <p>(1) Nerve (2) Choroid plexus</p> <p>(3) Neurons (4) Neuroglial cells</p> <p>157. Respiration, heart beat and peristalsis are controlled by-</p> <p>(1) Medulla oblongata</p> <p>(2) Medulla oblongata and cerebellum</p> <p>(3) Medulla oblongata and cerebrum</p> <p>(4) Cerebellum</p> <p>158. Which cells do not divide after birth-</p> <p>(1) Nerve cells (2) Germ cells</p> <p>(3) Muscle cells (4) Osteoblast</p> <p>159. Nerve impulse travels as-</p> <p>(1) Chemical impulse</p> <p>(2) Electrical impulse</p> <p>(3) Electrochemical impulse</p> <p>(4) Physical impulse</p> <p>160. Arbor vitae is composed of-</p> <p>(1) Grey matter (2) White matter</p> <p>(3) Neuroglia cells (4) All the above</p> | <p>161. The most common synapse in our body is</p> <p>(1) Electrical</p> <p>(2) Chemical</p> <p>(3) Both are equally present</p> <p>(4) None of these</p> <p>162. Medulla oblongata encloses-</p> <p>(1) Fourth ventricle (2) Third ventricle</p> <p>(3) Second ventricle (4) Optic lobes</p> <p>163. Fourth cranial nerve is-</p> <p>(1) Trigeminal (2) Abducens</p> <p>(3) Trochlear (4) Optic</p> <p>164. Sympathetic nervous system increases-</p> <p>(1) Heart beat</p> <p>(2) Secretion of saliva</p> <p>(3) Secretion of digestive juices</p> <p>(4) All the above</p> <p>165. Main function of cerebellum is-</p> <p>(1) Balancing (2) Sight</p> <p>(3) Hearing (4) Remembering</p> <p>166. Centre for sense of smell is-</p> <p>(1) Cerebellum (2) Olfactory lobes</p> <p>(3) Cerebrum (4) Midbrain</p> <p>167. Nerve related to diaphragm is-</p> <p>(1) Vagus</p> <p>(2) Phrenic</p> <p>(3) Trigeminal</p> <p>(4) Glossopharyngeal</p> <p>168. Which of the following is the longest nerve-</p> <p>(1) Vagus</p> <p>(2) Olfactory</p> <p>(3) Hypoglossal</p> <p>(4) Glossopharyngeal</p> <p>169. In an axon, nerve impulse travels-</p> <p>(1) Away from cyton</p> <p>(2) Towards cyton</p> <p>(3) Both away and towards cyton</p> <p>(4) Not known</p> <p>170. The junction between axon of one neuron and dendrites of the next neuron is called-</p> <p>(1) junction point (2) A joint</p> <p>(3) A synapse (4) Constant</p> |
|---|--|

171. Consider the following feature :

- (A) Exclusively marine in nature
 - (B) Contains extra and intra cellular digestion
 - (C) Only sexual reproduction occurs
 - (D) Development indirect
- Which of the following animal has all the above mentioned feature :

- (1) *Cliona*
- (2) *Pennatula*
- (3) *Pleurobrachia*
- (4) *Adamsia*

172. Coelom is a cavity present :-

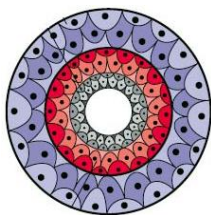
- (1) Between mesoderm layers
- (2) Between body wall and ectoderm
- (3) Between mesoderm and endoderm
- (4) Between mesoderm and body wall

173. Which of the following statement is not correct regarding the given figure :



- (1) Umbrella shaped and free swimming form
- (2) Dominates in life cycle of Aurelia
- (3) Produces polyp sexually in obelia
- (4) Shows alteration with polyp in adamsia

174. This diagram represent the organisation of :

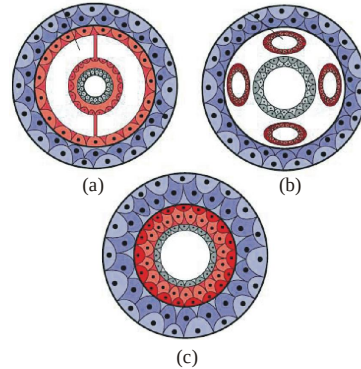


- (1) *Aurelia*
- (2) *Ascaris*
- (3) *Spongilla*
- (4) *Ctenoplane*

175. Which of the following animal has calcareous exoskeleton ?

- (1) *Leucosolenia*
- (2) *Obelia*
- (3) *Gorgonia*
- (4) *Physalia*

176. In the given diagram identify the coelomic condition in the animals :



- (1) (a) *Taenia* (b) *Ascaris* (c) Leech
- (2) (a) Scorpion (b) *Ancylostoma* (c) *Fasciola*
- (3) (a) Roundworm (b) Tapeworm (c) Earthworm
- (4) (a) Earthworm (b) *Schistosoma* (c) *Ascaris*

177. Gemmules are helpful in :

- (1) Digestion
- (2) Sexual reproduction
- (3) Secretion of spicules
- (4) Survival in drought

178. Nematocytes are the organ of :

- (1) Sensation
- (2) Reproduction
- (3) Defence
- (4) Digestion

179. Sea anemone is :

- (1) Diploblastic, Radiata
- (2) Diploblastic, Bilateral
- (3) Triploblastic, Radiata
- (4) Triploblastic, Bilateral

180. What will happen if cnidocil is touched with a glassrod ?

- (1) Nematoblast will be discharged
- (2) Nematoblast will not discharged
- (3) Nematoblast will die
- (4) No effect at all

Space for rough work

Space for rough work
